

COVETED TALENT (Left) Izamam Majgul, an athlete from the Siddi community, with his coach in Junagadh, Jan. 6, 2021

AFP

▶ GUJARAT

THE GENE HUNT SPORT

Sports genomics takes off in a big way in the state, with a big DNA data hunt to help isolate 'the athletic gene' among its peoples. Final aim: readiness for Olympics

Sports genomics is an emerging field that explores the relationship between genes and athletic potential, performance and training responses. Most global research is built on non-Indian populations; Gujarat's is a step to indigenise data. "We hope to gather 2,000 samples every year, starting 2026," says Dr Snehal Bagatharia, director, GBRC.

On an average, genetic factors are said to "contribute about 66 per cent to athletic performance". (The rest is environmental variables: training, nutrition, ergogenic aids, medical/ social support et al.) Patterns of DNA polymorphisms can help gauge endurance, strength, power and team sport suitability. Personalised training, optimising performance, minimising health risks, identifying talent early in appropriate social pools, it's all part of themix. "It's a powerful tool for evidence-based talent identification and precision training," says Bagatharia. With an annual budget of Rs 5.21 crore for the SGP, Gujarat also hopes to diversify its sports culture beyond cricket and position itself as a sports biotech hub.

Takeaways

- **Sports Genomics Programme to sequence DNA of 10,000 athletes**
- **DNA studies can aid performance, training, injuries, talent hunts**
- **Gujarat aims to be a sports biotech hub while aiming for Olympics**

Table tennis champion Sharath Kamal used genetic screening before Paris 2024 not to peak higher, but to age smarter. The BCCI brought genetic testing into the Indian men's setup in 2017, using it to fine-tune speed, endurance, fat metabolism, muscle development and recovery. These tests were introduced in India around 2011, with many players individually getting these assessments done before that.

There are also troubling questions. Who owns your DNA data? Could genetic profiling end up excluding talent from allegedly 'lesser' pools? Will labs replace grassroots development? Will the future go towards gene-editing? ■